

index 0. In that case, you simply need to remove the row at index 1.

3. Convert all the items in the table into numerics.

The script for implementing Step 2 is as follows:

```
# ----- STEP 2:- Clean the data
# Step 2(1) -- Change the column names
df2.columns = ['X1', 'Y1', 'X2', 'Y2',
               'X3', 'Y3', 'X4', 'Y4']
# print(df2) ## Uncomment to check
# that column names changed
# Step 2(2) -- Remove row containing x
# and y at index 0
df2.drop([0], axis = 0, inplace = True)
# print(df2) ## Uncomment to confirm
# that row with x and y removed
# Step 2(3) -- convert all columns of
# DataFrame to numeric values
df2 = df2.apply(pd.to_numeric)
# print (df2.dtypes) # Uncomment
# to see all cells of DataFrame have
# numerics
```

Step 3: Plot the four data sets

For plotting the four data sets, we will use the Matplotlib library. As you may be aware, the pyplot module of the matplotlib library has a function called `subplots()`. The signature is as follows:

```
matplotlib.pyplot.subplots(nrows=1,
                           ncols=1, sharex=False, sharey=False,
                           squeeze=True, subplot_kw=None,
                           gridspec_kw=None, **fig_kw)
```

You can see the complete signature of the function along with a description of all the parameters at the link given at the end of the article.

The two important parameters are `nrows` and `ncols`. Since we need four sub-plots in a 2x2 grid, we will give values `nrows = 2` and `ncols = 2`. You can ignore the two additional arguments, namely, `sharex` and `sharey`. Those who are interested can see the

example code in the link given under 'References' at the end of the article.

The script for implementing Step 3 is as follows:

```
# ----- STEP 3:- Plot the 4 data sets
%matplotlib inline
import matplotlib.pyplot as plt

# --- Step 3(1): Create 4 subplots
fig, ax = plt.subplots(nrows=2, ncols=2,
                      figsize = (30, 30),
                      sharex = 'col', sharey = 'row')

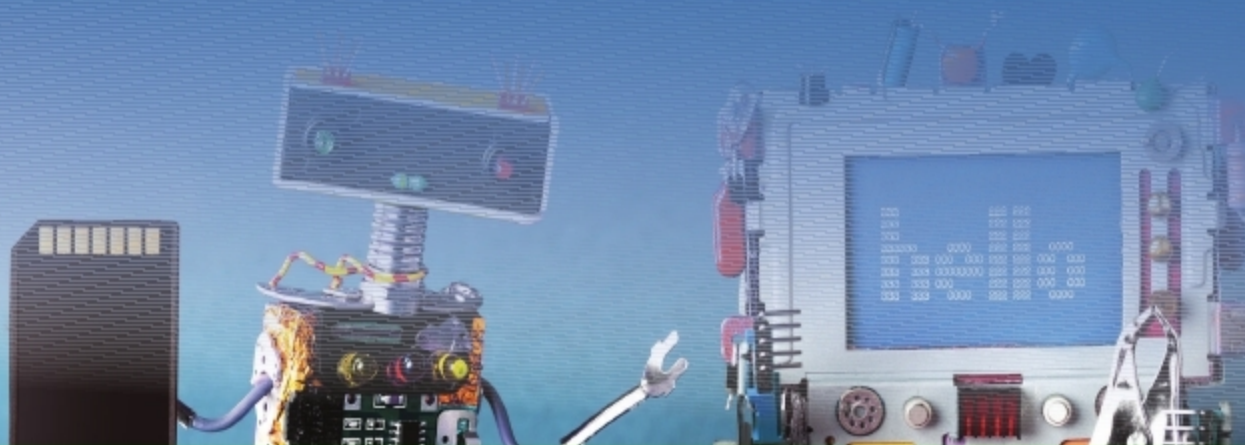
# --- Step 3(2):- Create 4 Scatter
# Plots
ax[0][0].scatter(df2.X1, df2.Y1, s =
                200)
ax[0][1].scatter(df2.X2, df2.Y2, s =
                200)
ax[1][0].scatter(df2.X3, df2.Y3, s =
                200)
ax[1][1].scatter(df2.X4, df2.Y4, s =
                200)
```

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